

COSC264 · INTRODUCTION TO COMPUTER NETWORKS AND THE INTERNET · TERM 4 · MODULE 6

Application Protocols: the Web, Email and DNS

1 · Network applications

2 · The Web and HTTP

3 · Email

4 · DNS

Module 6 – Application Protocols

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OUTLINE

Module 6 — roadmap

General concepts about network applications

One network application: the Web

HTTP: the protocol that supports the Web

Email

DNS: the Domain Name System

MODULE 6

Network Applications

What an application needs from the layers beneath it

1 • Network applications

2 • The Web and HTTP

3 • Email

4 • DNS

What is a network application?

Programs that run on different end systems and communicate with each other over the network.

Web browsing

Google Chrome, Mozilla Firefox,
Microsoft Edge

Email

Gmail, Outlook

Instant messaging

Slack, WhatsApp, Facebook
Messenger

Streaming services

YouTube, Netflix

Social media

Instagram, TikTok

AI assistants

Claude, ChatGPT, Gemini

Application-layer protocols

An **application-layer protocol** is one piece of a network application — but a big piece.

It defines how the application's processes, running on different end systems, **pass messages to each other**: which messages exist, their format, and when and how to respond.

This is what lets programs on different machines work together — e.g. any web browser can fetch a page from any web server, whoever wrote each program.

Network application	Application-layer protocol
Web application	HTTP
Email	SMTP

Choosing a transport protocol

- Application-layer protocols are controlled by the **application developer**; transport-layer protocols, however, are built into the **OS**.
- As a developer you can only choose between the transport protocols the OS provides — so pick the one that best fits what your application-layer protocol needs.

Property	Example applications	Transport protocol
Reliable data transfer	Email, IM, file transfer, financial apps	TCP preferred UDP unsuitable
Low delay	Internet telephony, conferencing, multiplayer games	UDP preferred TCP retransmissions add latency

MODULE 6

The Web and HTTP

The application-layer protocol behind every page you load

1 · Network applications

2 · **The Web and HTTP**

3 · Email

4 · DNS

The World Wide Web

THE WEB

The **World Wide Web** (aka Web) is a **network application**: a global collection of documents and other resources shared over the Internet — like a large file storage system.

Each object on the Web (e.g., an image, an HTML file) is addressable by a **URL**:

```
http://www.someschool.edu/someDept/pic.gif  
└─ host name ─┘ └─ path name ─┘
```

HTTP

HTTP: the protocol behind the Web

HTTP

The **Hypertext Transfer Protocol** is the **application-layer protocol** that supports the Web — it defines how clients request objects by URL and how servers respond.

HTTP follows a **request–response** pattern: the client sends an **HTTP request** for an object, and the server answers with an **HTTP response** carrying it.



HTTP

How HTTP works

HTTP uses **TCP** as its transport protocol — so before any HTTP message is sent, a TCP connection must be established first.



STATE LESS

HTTP is **stateless** — the server maintains no information about past client requests. Each request is handled independently.

HTTP connections

Non-persistent HTTP

- At most **one object** is sent over a TCP connection — connection is then closed
- HTTP/1.0 uses non-persistent HTTP

Persistent HTTP

- **Multiple objects** can be sent over a single TCP connection
- HTTP/1.1, HTTP/2, and HTTP/3 use persistent HTTP

Protocol	Published	Connection type	Status
HTTP/1.0	1996	Non-persistent	Legacy
HTTP/1.1	1997	Persistent	Standard
HTTP/2	2015	Persistent	Standard
HTTP/3	2022	Persistent (QUIC)	Standard

Non-persistent HTTP in action

Suppose a client wants to retrieve a file from: `http://www.someSchool.edu/someDepartment/home.index`

1. Client establishes a **TCP connection** with the server
2. Client sends an **HTTP GET** request — path: `/someDepartment/home.index`
3. Server retrieves the HTML file, wraps it in an **HTTP response**, and sends it back
4. Server signals TCP to **close the connection**



Response time: non-persistent HTTP

RESPONSE TIME

The time to retrieve a single object using HTTP — from the client's request until the object is received

Q: What is the response time for non-persistent HTTP?

A: 2 RTTs + file transmission time

RTT

Recall: **round-trip time** — the time for a small packet to travel from client to server and back

First RTT: the TCP handshake — client sends SYN, server replies SYN-ACK

Second RTT: client sends HTTP GET, server sends back the response

With many referenced objects, this cost multiplies — a significant burden on the server



Persistent HTTP in action

Client and server establish a TCP connection and exchange HTTP messages

Server **keeps the TCP connection open** after sending the first HTTP response

Subsequent HTTP messages between the same client and server use the **same TCP connection** — no new handshake needed

Response time on one persistent connection:

1 RTT for the initial TCP handshake, then 1 RTT for each HTTP request.

So for n requests: **total** $\approx (1 + n)$ RTT (plus transmission time).

Amortised over the n objects: $(1 + n)/n = (1/n + 1)$ RTT. As n becomes large, $1/n \rightarrow 0$, so this approaches **~1 RTT per object**.

Can we do even better than 1 RTT per object?



HTTP

Improving HTTP efficiency further with pipelining

**WITHOUT
PIPELINING**

Client issues a new request **only after** receiving the previous response —
1 RTT per file



Improving HTTP efficiency further with pipelining

WITHOUT PIPELINING

Client issues a new request **only after** receiving the previous response — **1 RTT per file**

WITH PIPELINING

Client sends requests **as soon as it needs to retrieve an object**, without waiting for responses — e.g. in this example we pipeline 3 HTTP requests all together

Benefit: pipelining 3 requests → response time per file $\approx 1 \text{ RTT} / 3$

- Don't confuse this with TCP-level pipelining.
- **HTTP pipelining** operates at the **application layer**: the client sends several **HTTP requests** without waiting for each response.
- **TCP pipelining** operates at the **transport layer**: the sender keeps several **segments** in flight within its window. TCP cuts each HTTP message into segments, so these may be pieces of one HTTP request, or of several.



HTTP request message format

Method — what action to perform

Method	Action
GET	Retrieve a resource
HEAD	Retrieve headers only (no body)
POST	Submit data to a resource
PUT	Store / replace a resource
DELETE	Remove a resource

Request line

method URL version cr lf

Header lines

field name: value cr lf
field name: value cr lf

Blank line

cr lf

Entity body

entity body

HTTP request message format

URL: path to the requested resource — e.g.

`/someDept/pic.gif`

HTTP Version: identifies the protocol in use — HTTP/1.0 or HTTP/1.1

Request line

method

URL

version

cr lf

Header lines

field name:

value

cr lf

field name:

value

cr lf

Blank line

cr lf

Entity body

entity body

HTTP request message format

Header lines

- Zero or more `field-name: value` pairs, one per line
- Common examples:

```
Host: www.example.com
```

Mandatory in HTTP/1.1 — identifies the target server

```
Content-Type: text/html; charset=UTF-8
```

Format of the message body

```
Connection: close
```

Close the TCP connection after this response

```
Accept-Language: da, en-gb;q=0.8,  
en;q=0.7
```

Preferred languages, weighted by q (default 1): Danish → British English → any English

Request line

`method``URL``version``cr lf`

Header lines

`field name:``value``cr lf``field name:``value``cr lf`

Blank line

`cr lf`

Entity body

entity body

HTTP

HTTP request message format

Entity body: empty for GET and HEAD; carries the payload for POST and PUT (e.g. form data or JSON)



HTTP

HTTP request: an example

```
GET /somedir/page.html HTTP/1.1
```

```
Host: www.someschool.edu
```

```
User-agent: Mozilla/4.0
```

```
Connection: close
```

```
Accept-language: fr
```

(empty entity body)

Method: GET

URL: /somedir/page.html

HTTP version: HTTP/1.1

Header field: Host: www.someschool.edu

Header field: User-agent: Mozilla/4.0

Header field: Connection: close

Header field: Accept-language: fr

Body: (empty)

HTTP response message format

Version

The HTTP version the server is using — e.g. HTTP/1.1, just as in the request line.

Status line

version

status code

phrase

cr lf

Header lines

field name:

value

cr lf

field name:

value

cr lf

Blank line

cr lf

Entity body

entity body

HTTP response message format

Status code & phrase

- **Status code:** 3-digit number indicating the result of the request
- **Phrase:** human-readable text accompanying the code — e.g. OK, Not Found

Status line

version — status code — phrase — cr lf

Header lines

field name: — value — cr lf

field name: — value — cr lf

Blank line

cr lf

Entity body

entity body

HTTP response message format

Common examples

Code	Phrase
200	OK
301	Moved Permanently
400	Bad Request
404	Not Found
505	HTTP Version Not Supported

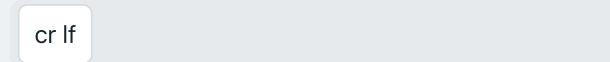
Status line



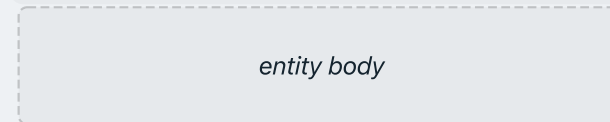
Header lines



Blank line



Entity body



HTTP response message format

Header lines, blank line, entity body

Same structure as in the request: `field-name: value`
header lines, a blank line (a lone CRLF), then the body —
here, the **requested object** itself.



HTTP response: an example

```
HTTP/1.1 200 OK
Connection: close
Date: Thu, 06 Aug 1998 12:00:15 GMT
Server: Apache/1.3.0 (Unix)
Last-Modified: Mon, 22 Jun 1998 ...
Content-Length: 6821
Content-Type: text/html
```

```
<!DOCTYPE html>
<html><head><title>Example</title></head>
<body><h1>Hello!</h1>
<script>console.log("hi");</script>
</body></html>
```

Version: HTTP/1.1

Status code: 200 — success

Phrase: OK

Header: Connection: close — close after this response

Header: Date — when the response was sent

Header: Server — the server software

Header: Last-Modified — when the object last changed

Header: Content-Length — body size in bytes

Header: Content-Type — the body is HTML

Body: the requested HTML file (with embedded JS)

Cookies

HTTP is **stateless** — every request is independent; the server retains no memory of past requests.

So the server does not remember your earlier conversations with it. Yet many applications *need* that memory: when you refresh an online shop's page, the items in your cart should not just disappear. **How does the server know what is in your cart?**

The answer: **cookies** — here's how it works.

1. A **Set-Cookie** header line in the **HTTP response** — the server sends the client a unique ID on first contact
2. A **Cookie** header line in subsequent **HTTP requests** — the client echoes the ID back on every future request
3. A **cookie file** stored on the client, managed by the browser — persists across sessions
4. A **back-end database** on the server, keyed by the cookie ID — stores per-user state

Cookies: an example [RFC 6265]



The client sends an HTTP request to the host

Set-Cookie: id=1678 — Server assigns a unique ID and sends it to the client

The cookie ID is stored as an entry in the back-end database, and in the client's browser

Cookies: an example [RFC 6265]



Cookie: id=1678 — Client attaches the ID on every future request to the same server

Server uses the ID to look up the user's state in its database

Server can then generate a **tailored response** — e.g. a personalised page, a pre-filled shopping cart — for the client's request

HTTP

One problem with the Web

Picture a university: thousands of users fetch the same popular pages and images, and every request travels across the Internet to the **origin server** that hosts the object.

Every one of those requests lands on the same origin server, through the same congested links — enough of them can **overwhelm** it.

How can we avoid this?

Web caching

A **web cache** (also called a **proxy server**) is a network entity that satisfies HTTP requests on behalf of an origin server — storing copies of recently requested objects locally.



Acts as a server

Receives HTTP requests from clients and, if it holds a copy of the object, returns it in the response — just like an origin server would

Acts as a client

When the object is not cached, it issues its own HTTP request to the origin server, stores the copy, then answers the client

Requests answered by the cache get a **quicker response**, and caching **reduces how often** requests have to go to the origin server.

Conditional GET



A **conditional GET** is an ordinary GET with an `If-modified-since` header line. A proxy uses it to check whether its cached copy is still up to date, e.g.:

```
GET /index.html HTTP/1.1
```

```
Host: www.example.com
```

```
If-modified-since: Mon, 10 Aug 2026 09:00:00  
GMT
```

If the object is **unchanged**, server replies:

```
304 Not Modified
```

Object **changed**: server replies 200 OK with the updated object in the body

MODULE 6

Email

Electronic Mail

1 · Network applications

2 · The Web and HTTP

3 · Email

4 · DNS

An overview of the email system



User agent (UA) — the app used to read, compose and send email (e.g. Outlook, Gmail, Apple Mail)

Mail server — a dedicated host that handles email on behalf of its users. It stores a **mailbox** per recipient for incoming mails and a **send queue** of outgoing mails

SMTP — Simple Mail Transfer Protocol; the sender's UA uses it to send a message to its own mail server

The sender's mail server then uses **SMTP** to transfer the message to the recipient's mail server across the internet

Mail access protocol (MAP) — used by UAs to retrieve messages from their mail server

SMTP in action

SMTP has a **client side** and a **server side**, and always runs over **TCP**.



① Alice uses her UA to **compose** the message

② Alice's UA (**SMTP client**) **sends** the message to her mail server (**SMTP server**)

③ The **SMTP client** on Alice's server opens a TCP connection to Bob's mail server and **transfers** Alice's message over it

④ The **SMTP server** on Bob's server receives the message and places it in Bob's **mailbox**

Retrieving email from the server



The recipient's UA retrieves email from its mail server either with a **mail access protocol (MAP)**, e.g.

- **POP3** — Post Office Protocol version 3 [RFC 1939]
- **IMAP** — Internet Message Access Protocol [RFC 9051]

or through **web-based email**, where the UA is a web browser that uses **HTTP** to reach the mail server

How POP3 works



IMAP

Internet Message Access Protocol

IMAP addresses POP3's limitations:

IMAP **keeps per-message state on the server**, e.g. read or unread, flagged, so every device sees the same status

It supports **folders on the server**: create folders and move messages into them, visible from any device

Accessing email via a browser

Web-based email is accessed through a web browser

The mailbox stays on the server; reading an email is like browsing the Web — the browser fetches each message as a web page

Which application-layer protocol does the browser use to talk to the mail server?

HTTP, the same protocol behind every other web page

Web-based email services

Examples: **Gmail**, **Outlook** (web)

Comparing POP3, IMAP and web-based email

	POP3	IMAP	Web-based email
Accessed with	A mail app	A mail app	A web browser, over HTTP
Messages	Downloaded to the device; deleted from the server, or optionally kept	Kept on the server	Kept on the server
Folders	None on the server; a single mailbox	Created and kept on the server	Created and kept on the server
Multiple devices	Not synced: read status and folders exist only on each device	Read status and folders stay in sync across devices	Any device with a browser sees the same mailbox

Structure of an email message

An email message consists of **header fields**, followed (optionally) by a **body**

Header fields describe the message, e.g. **From:** and **Subject:**

The **body** carries the message content itself

The whole message is **ASCII text**, which encodes English letters, digits and punctuation

Problem: ASCII alone cannot carry images, attachments, or non-ASCII characters (e.g. Chinese, Arabic)

Date: Mon, 10 Aug 2026 09:00 GMT

From: alice@crepes.fr

To: bob@hamburger.edu

Subject: Picture of yummy crepe.

Cc: charlie@example.com

Header fields

Hello Bob, here is the picture
you asked for. Enjoy!

Body

Multipurpose Internet Mail Extensions

MIME extends the message format with **additional header fields**

These fields declare the content's **type** and how it is **encoded**

so the receiver knows how to **decode the body**

e.g. **base64** encoding converts binary data into printable ASCII text, which can be used for image attachments

From: alice@crepes.fr

To: bob@hamburger.edu

Subject: Picture of yummy crepe.

MIME-Version: 1.0

Content-Transfer-Encoding: base64

Content-Type: image/jpeg

(base64 encoded data...
...base64 encoded data)



decoded image

MODULE 6

DNS

Domain Name System

1 · Network applications

2 · The Web and HTTP

3 · Email

4 · **DNS**

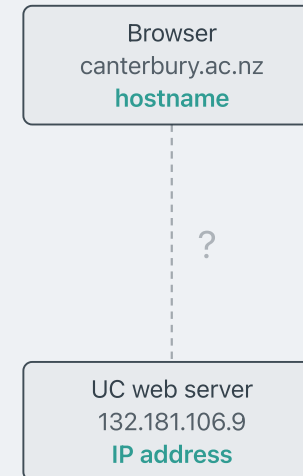
Hostnames and IP addresses

- As we have seen, network routing entirely relies on **IP addresses** — numerical identifiers, e.g. `132.181.106.9`
- However, when you browse the web, you typically type a **human-friendly hostname** such as `canterbury.ac.nz`
- How does the browser know how to translate between a hostname and an IP address?



DNS

Domain Name System — overview



Domain Name System — overview

DNS is that translation mechanism — it maps human-readable **hostnames** to numerical **IP addresses**

DNS stores this mapping in a **distributed database**, spread across a hierarchy of DNS servers [RFC 1034, 1035]

DNS is an **application-layer protocol** running mainly over **UDP on port 53** (TCP port 53 for large responses and zone transfers)

Why UDP? A query and its reply each fit in one small datagram, so there is no connection setup delay, providing a quick response.



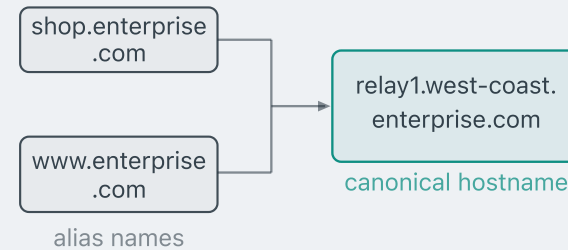
Additional services provided by DNS

Hostname-to-IP translation — the primary service, already seen; commonly used by HTTP, SMTP, and other application-layer protocols

Host aliasing — a host with a complex **canonical hostname** can be reached via simpler **alias names**

e.g. `www.enterprise.com` and `shop.enterprise.com` both resolve to the canonical `relay1.west-coast.enterprise.com`

Mail server aliasing — the same idea for email; a friendly address like `bob@enterprise.com` can map (via an MX record) to the canonical mail server hostname `relay1.west-coast.enterprise.com`

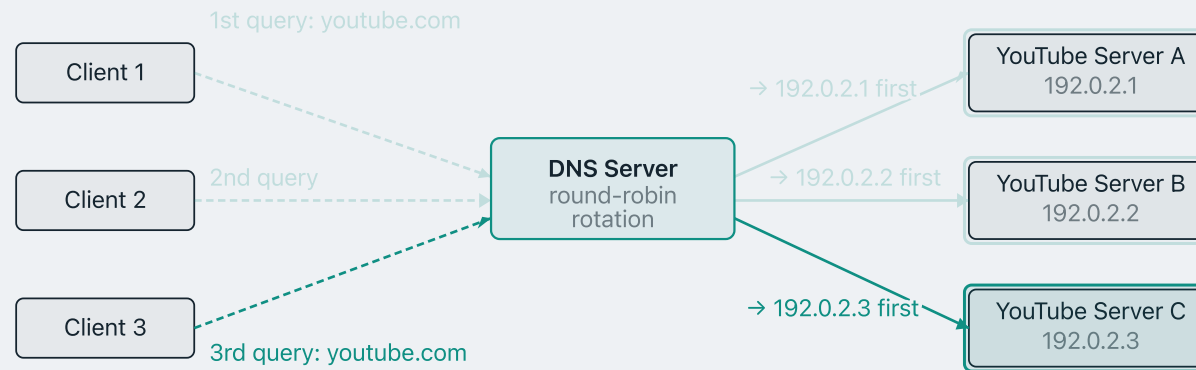


Additional services by DNS: load distribution

Many popular web services — e.g. YouTube, Google, Netflix — are hit by **millions of users** simultaneously. A **single server** cannot handle that volume.

The service is therefore distributed across **many server instances**, each with a **distinct IP address**, all sharing the **same hostname** (e.g. `youtube.com`).

How are users assigned to these servers? **DNS load balancing** — DNS stores all the IPs and **rotates their order** on each response, so consecutive clients get a different IP listed first.



Why not a centralised design?

A centralised design would use a **single server** that hosts the translation database — every DNS query goes to that server.

Single point of failure

If the one server crashes, DNS lookup fails for the entire Internet

Traffic volume

Billions of DNS queries per day — no single machine can handle that load

Distant centralised database

All queries must travel to one location — users far away face high latency on every lookup

Maintenance

Billions of hostname records change constantly — impossible to keep one database current

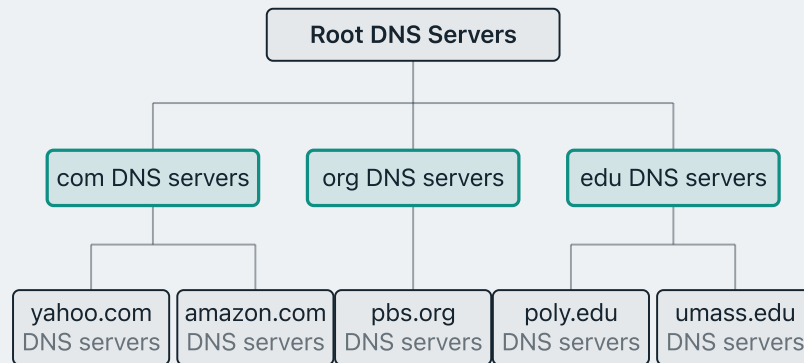
A hierarchical distributed database



Root DNS Servers

13 root server identities (a–m.root-servers.net) — each replicated via anycast, giving **over 1000 physical instances** worldwide

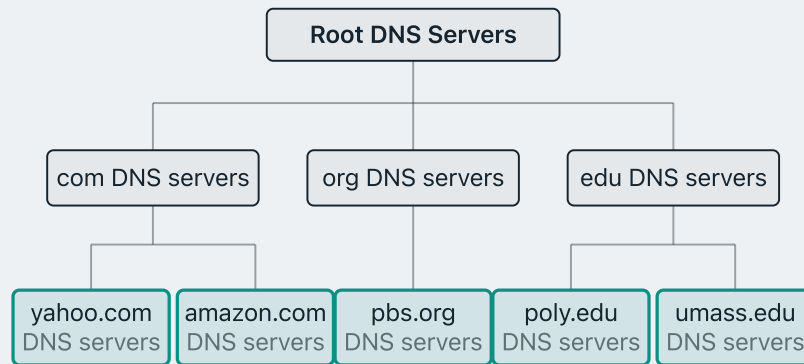
A hierarchical distributed database



TLD DNS Servers

Handle **.com**, **.org**, **.edu** and country-code domains (.nz, .uk). Maintained by dedicated registries — e.g. **Verisign** (.com), **Public Interest Registry** (.org), **EDUCAUSE** (.edu), **InternetNZ** (.nz).

A hierarchical distributed database



Authoritative DNS Servers

Each organisation's own DNS server — the **definitive source** for its hostname-to-IP mappings.
e.g. Canterbury's authoritative server holds the IP for `www.canterbury.ac.nz`

Local DNS server and DNS caching

Every ISP or institution operates a **local DNS server** (also called a *default name server*). It is **not** part of the root / TLD / authoritative hierarchy — it sits **separately** from that.

When you join the network, your device is automatically configured to use one — e.g. the University of Canterbury runs its own local DNS server for all campus users.

DNS Hierarchy
Root · TLD · Authoritative
(separate from local DNS)

Local DNS server
e.g. dns.canterbury.ac.nz
(cache)

Your computer

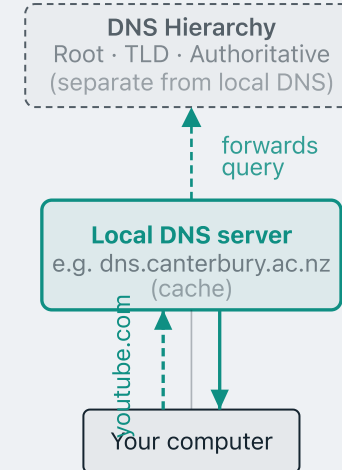
Local DNS server and DNS caching

When your machine makes a DNS lookup, the query goes **first to the local DNS server**.

If the answer is cached: the local DNS server replies **immediately** — no hierarchy lookup needed.

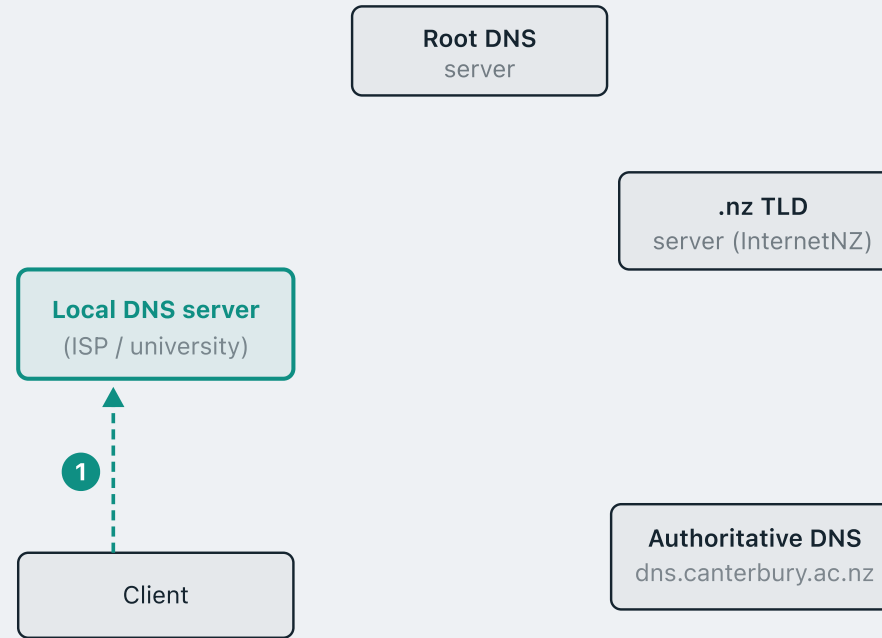
If not cached: the local server **forwards the query into the hierarchy**, relays the reply back to you, and **caches the hostname-to-IP mapping** in local memory for future queries.

Cache entries have a **TTL (time-to-live)** and expire after some time (typically hours). TLD server addresses are routinely cached in local DNS servers — so **root servers are bypassed for most everyday queries**.



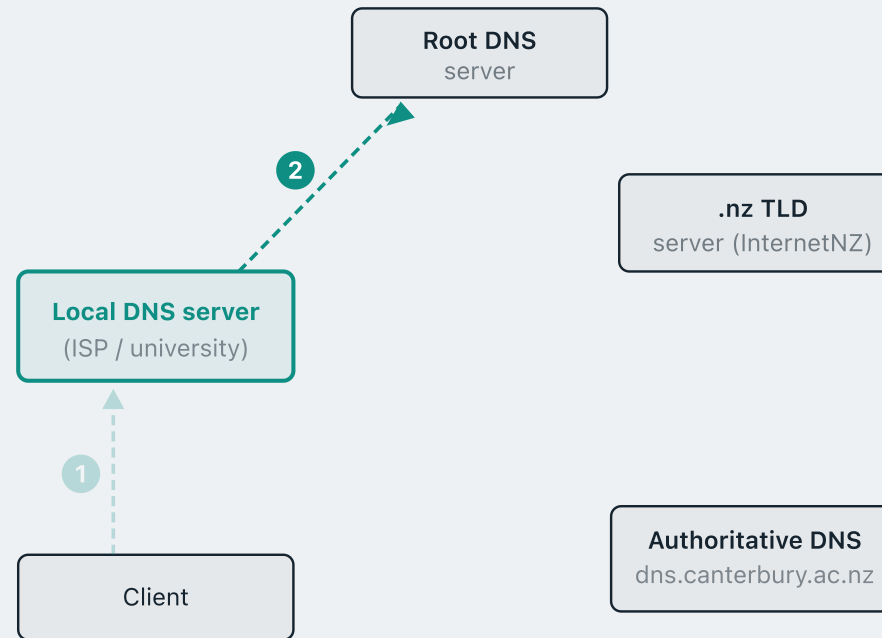
Resolving a hostname: the full journey

Step 1. A client wants to resolve **csse.canterbury.ac.nz**. It sends a DNS query to its **local DNS server**.



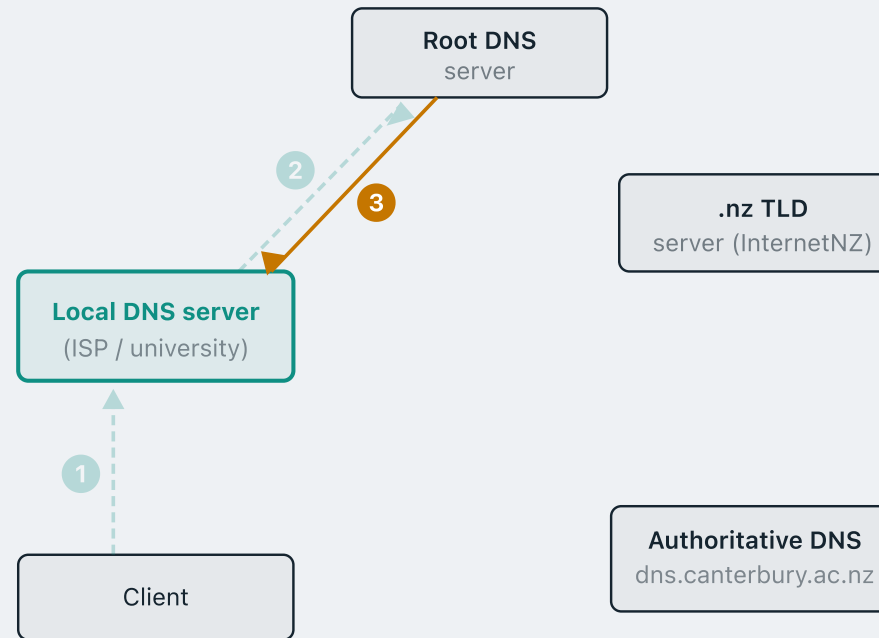
Resolving a hostname: the full journey

Step 2. The local DNS doesn't have the answer cached. It contacts a **root DNS server**.



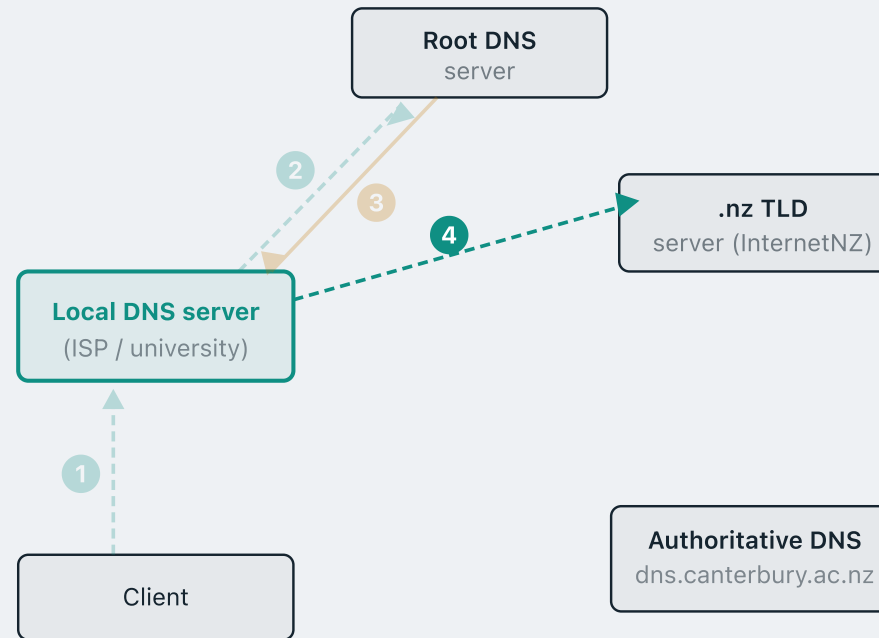
Resolving a hostname: the full journey

Step 3. The root server returns a **referral**: "I don't know — try a **.nz TLD** server" and gives its address.



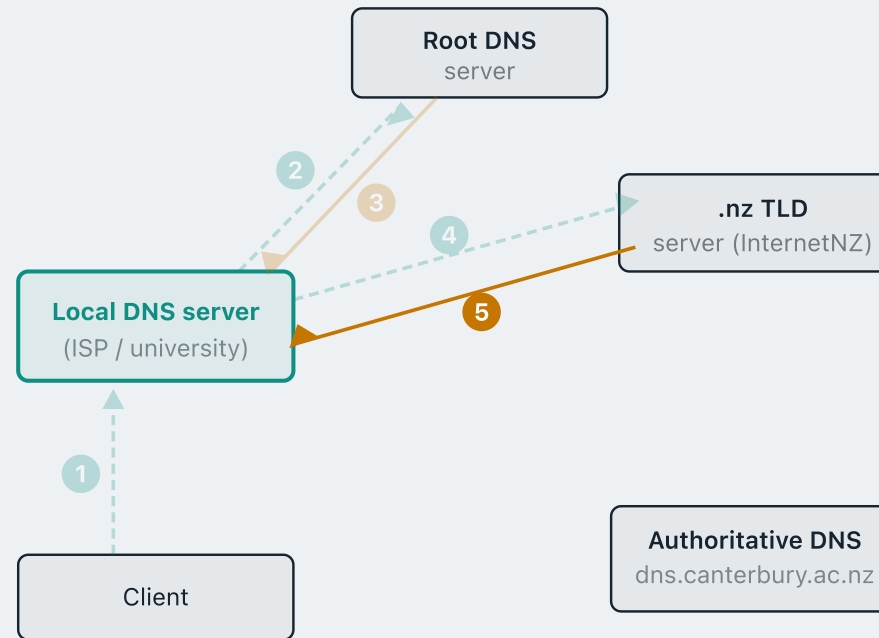
Resolving a hostname: the full journey

Step 4. The local DNS follows the referral and contacts the **.nz TLD server**.



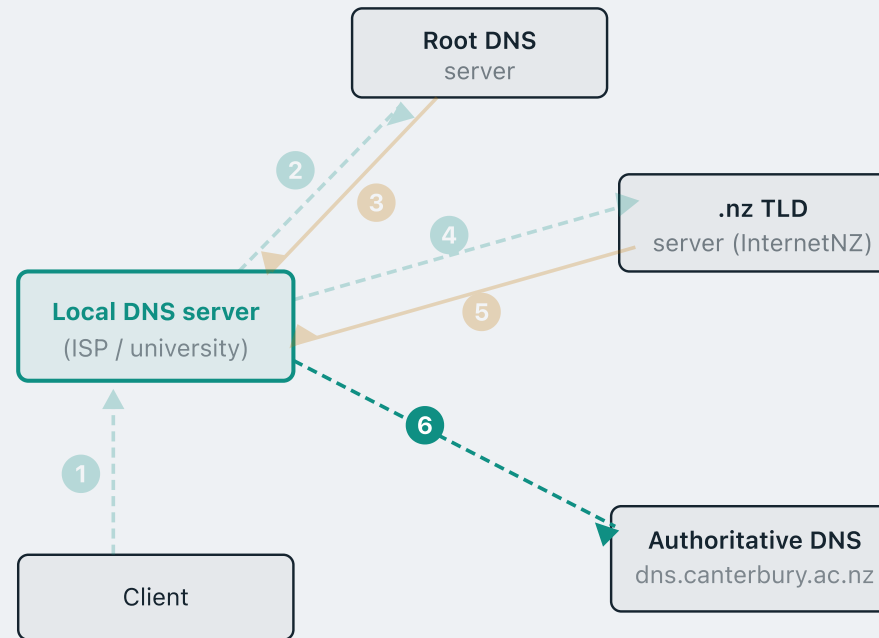
Resolving a hostname: the full journey

Step 5. The .nz TLD returns another **referral**:
"Try **dns.canterbury.ac.nz** — the
authoritative server for canterbury.ac.nz."



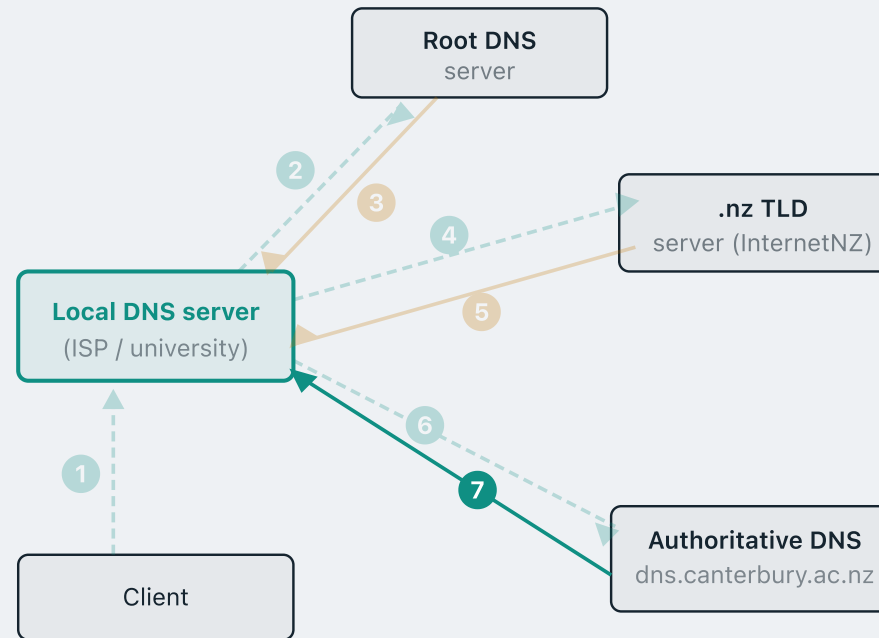
Resolving a hostname: the full journey

Step 6. The local DNS contacts **dns.canterbury.ac.nz** — the authoritative name server for the canterbury.ac.nz zone.



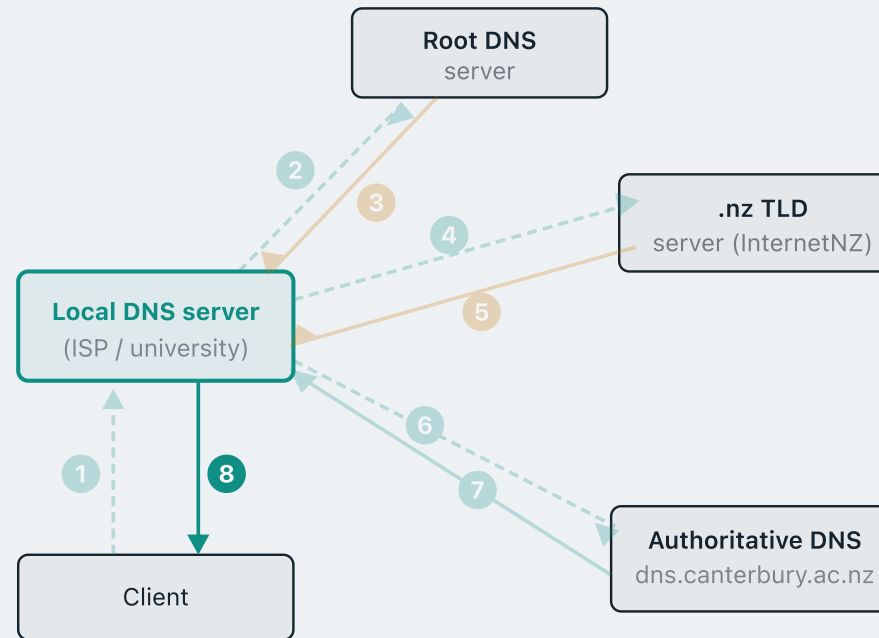
Resolving a hostname: the full journey

Step 7. The authoritative server has the answer! It replies: "csse.canterbury.ac.nz = 132.181.2.21"



Resolving a hostname: the full journey

Step 8. The local DNS returns **132.181.2.21** to the client — and **caches** the mapping for future queries.



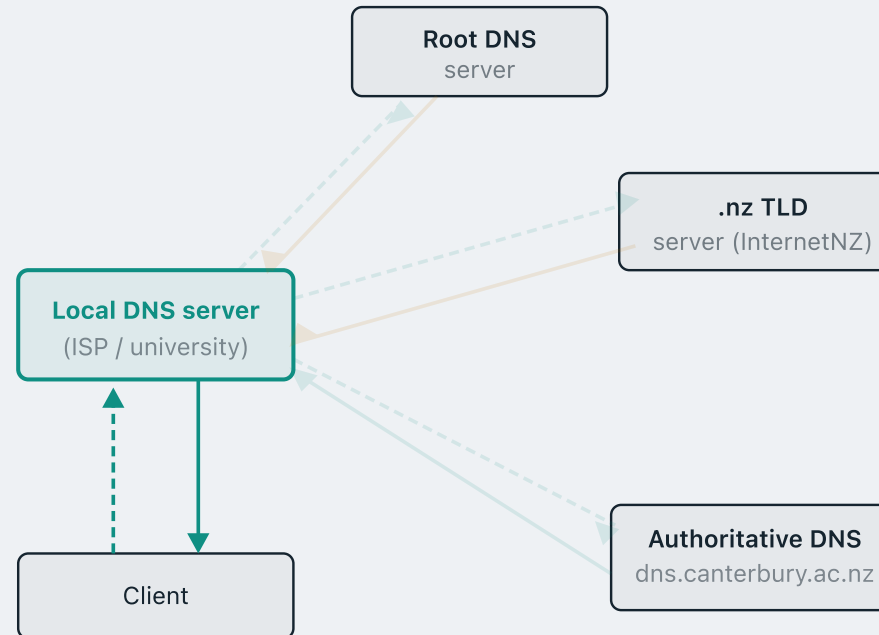
Recursive vs iterative queries

Iterative query

The contacted server does **not** resolve the name for you. It returns a **referral** — "I don't know, try that server."

Recursive query

The contacted server takes **full responsibility** — it resolves the name itself (querying the next server in the chain) and returns the **final answer**.



Recursive vs iterative queries

Iterative query

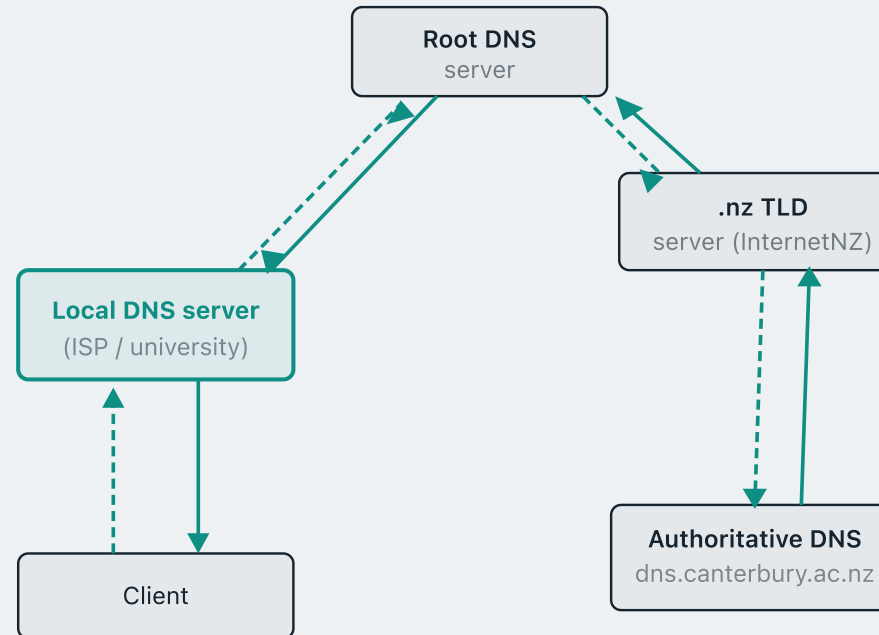
The contacted server does **not** resolve the name for you. It returns a **referral** — "I don't know, try that server."

Recursive query

The contacted server takes **full responsibility** — it resolves the name itself (querying the next server in the chain) and returns the **final answer**.

Can we go fully recursive?

In principle, yes — root asks TLD, which asks auth. In practice, root and TLD servers refuse recursion (too much load), so only client → local DNS is recursive.



DNS Resource Records

DNS distributes data as **Resource Records (RRs)**. Each record has four fields:

(name, value, type, TTL – Time-To-Live)

NAME	VALUE	TYPE	TTL
web.canterbury.ac.nz	132.181.106.9	A	XXXX
itdns.canterbury.ac.nz	132.181.2.225	A	XXXX
mail.canterbury.ac.nz	132.181.3.225	A	XXXX
canterbury.ac.nz	itdns.canterbury.ac.nz	NS	XXXX
www.canterbury.ac.nz	web.canterbury.ac.nz	CNAME	XXXX
canterbury.ac.nz	mail.canterbury.ac.nz	MX	XXXX

DNS Resource Records

Type A

name = hostname · **value** = IPv4 address

NAME	VALUE	TYPE	TTL
web.canterbury.ac.nz	132.181.106.9	A	XXXX
itdns.canterbury.ac.nz	132.181.2.225	A	XXXX
mail.canterbury.ac.nz	132.181.3.225	A	XXXX
canterbury.ac.nz	itdns.canterbury.ac.nz	NS	XXXX
www.canterbury.ac.nz	web.canterbury.ac.nz	CNAME	XXXX
canterbury.ac.nz	mail.canterbury.ac.nz	MX	XXXX

DNS Resource Records

Type NS

name = domain · **value** = hostname of the authoritative DNS server for that domain

"I don't know — but here's the name of the server that does."

NAME	VALUE	TYPE	TTL
web.canterbury.ac.nz	132.181.106.9	A	XXXX
itdns.canterbury.ac.nz	132.181.2.225	A	XXXX
mail.canterbury.ac.nz	132.181.3.225	A	XXXX
canterbury.ac.nz	itdns.canterbury.ac.nz	NS	XXXX
www.canterbury.ac.nz	web.canterbury.ac.nz	CNAME	XXXX
canterbury.ac.nz	mail.canterbury.ac.nz	MX	XXXX

DNS Resource Records

Type CNAME

name = alias hostname · **value** = canonical (real) hostname it maps to

NAME	VALUE	TYPE	TTL
web.canterbury.ac.nz	132.181.106.9	A	XXXX
itdns.canterbury.ac.nz	132.181.2.225	A	XXXX
mail.canterbury.ac.nz	132.181.3.225	A	XXXX
canterbury.ac.nz	itdns.canterbury.ac.nz	NS	XXXX
www.canterbury.ac.nz	web.canterbury.ac.nz	CNAME	XXXX
canterbury.ac.nz	mail.canterbury.ac.nz	MX	XXXX

DNS Resource Records

Type MX

name = domain · **value** = hostname of the mail server for that domain

NAME	VALUE	TYPE	TTL
web.canterbury.ac.nz	132.181.106.9	A	XXXX
itdns.canterbury.ac.nz	132.181.2.225	A	XXXX
mail.canterbury.ac.nz	132.181.3.225	A	XXXX
canterbury.ac.nz	itdns.canterbury.ac.nz	NS	XXXX
www.canterbury.ac.nz	web.canterbury.ac.nz	CNAME	XXXX
canterbury.ac.nz	mail.canterbury.ac.nz	MX	XXXX

DNS protocol message format

DNS uses the **same message format** for both **queries** and **replies** — a fixed 12-byte header followed by four variable-length sections.

identification	flags	← 12 bytes
number of questions	number of answer RRs	
number of authority RRs	number of additional RRs	
questions variable number of questions		
answers variable number of resource records		
authority variable number of resource records		
additional information variable number of resource records		

DNS protocol message format

Identification

16-bit number assigned by the client. The reply copies this same value, so the client can match each reply to its original query.

identification	flags	← 12 bytes
number of questions	number of answer RRs	
number of authority RRs	number of additional RRs	
questions variable number of questions		
answers variable number of resource records		
authority variable number of resource records		
additional information variable number of resource records		

DNS protocol message format

Flags

- Query (0) or reply (1)
- Recursion desired — client requests full recursive resolution
- Recursion available — server indicates it supports recursion

identification	flags	← 12 bytes
number of questions	number of answer RRs	
number of authority RRs	number of additional RRs	
questions variable number of questions		
answers variable number of resource records		
authority variable number of resource records		
additional information variable number of resource records		

DNS protocol message format

Count fields

Four 16-bit counters — one per section below — telling how many records appear in each variable-length section.

identification	flags	← 12 bytes
number of questions	number of answer RRs	
number of authority RRs	number of additional RRs	
questions variable number of questions		
answers variable number of resource records		
authority variable number of resource records		
additional information variable number of resource records		

DNS protocol message format

Questions

The name and record type being queried. e.g. this
Type A question asks for the IP address of the domain
facebook.com.

(facebook.com, A)

identification	flags	← 12 bytes
number of questions	number of answer RRs	
number of authority RRs	number of additional RRs	
questions variable number of questions		
answers variable number of resource records		
authority variable number of resource records		
additional information variable number of resource records		

DNS protocol message format

Answers

Resource Records (RRs) that directly answer the query.

```
(facebook.com, 157.240.22.35, A,
TTL=300)
```

identification	flags	← 12 bytes
number of questions	number of answer RRs	
number of authority RRs	number of additional RRs	
questions variable number of questions		
answers variable number of resource records		
authority variable number of resource records		
additional information variable number of resource records		

DNS protocol message format

Authority

NS records pointing to authoritative servers —
returned in referral responses.

(facebook.com, a.ns.facebook.com, NS)

identification	flags	← 12 bytes
number of questions	number of answer RRs	
number of authority RRs	number of additional RRs	
questions variable number of questions		
answers variable number of resource records		
authority variable number of resource records		
additional information variable number of resource records		

DNS protocol message format

Additional

'Glue' Resource Records (RRs) accompanying the authority section — so you can reach the name server without an extra lookup.

```
(a.ns.facebook.com, 129.134.30.12, A)
```

identification	flags	← 12 bytes
number of questions	number of answer RRs	
number of authority RRs	number of additional RRs	
questions variable number of questions		
answers variable number of resource records		
authority variable number of resource records		
additional information variable number of resource records		

Query & response: a live example

QUERY →

Transaction ID 0x6492
Flags 0x0100 (query)
Questions 1 Answer RRs 0
Authority RRs 0 Additional RRs 0

QUESTIONS

www.facebook.com A

← RESPONSE

Transaction ID 0x6492
Flags 0x8180 (response, no error)
Questions 1 Answer RRs 2
Authority RRs 4 Additional RRs 8

QUESTIONS

www.facebook.com A

ANSWERS

www.facebook.com CNAME
star-mini.c10r.facebook.com
star-mini.c10r.facebook.com A
31.13.78.35

AUTHORITY

c10r.facebook.com NS
d.ns.c10r.facebook.com (+3)

ADDITIONAL

d.ns.c10r.facebook.com A
185.89.219.11 (+7)

DNS — key takeaways

- **Main purpose** — DNS maps human-readable hostnames to IP addresses.
- **Hierarchy** — DNS is distributed: no single server holds all records.
Responsibility is delegated across Root, TLD, and Authoritative servers.
- **Traversal pattern** — the client queries the local DNS resolver, which then follows referrals Root → TLD → Authoritative.
Results are cached locally by TTL to avoid repeated lookups.

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